ECON 895 Coding Exercise 1

Spatial Techniques in Empirical Economics

Distributed Monday 5 October Due Monday 19 October, 7:20 PM

Two halves, ten points each. The build half asks you to make something. The audit half asks you to find what is wrong with something a machine made. They are weighted equally.

AI use. AI is permitted and expected on both halves. Any tool is allowed. Every submission includes a short *AI use note* saying what you asked the tools to do and what you checked before signing off. You sign the result and you own every number and every line.

Submit by email to `njohnsoL@gmu.edu` by 7:20 PM on Monday 19 October, from your Mason address, with the subject line "Exercise 1". Attach three files named with your last name: `lastname_exercise1.R` for the build half, `lastname_audit` in any readable format for the audit half, and `lastname_ai_note` for the AI use note.

Your build script must run in a fresh R session without errors. Load your packages at the top, and read data from `data/` with paths relative to the repository root, as the weekly scripts do. I will run it.

Build half. Is Wittenberg an instrument?

Week 2 used distance from Mainz to instrument for early adoption of the printing press. Mainz is where Gutenberg worked, so the logic is that the technology spread outward from there.

Wittenberg is a different candidate. It is where Luther published the Ninety-five Theses in 1517, and Becker and Woessmann (2009, *Quarterly Journal of Economics*) use distance from Wittenberg to instrument for the spread of Protestantism.

The question. Does distance from Wittenberg work as an instrument for whether a city had a printing press by 1500, as Week 2 defined it?

1. Build `km_wittenberg` for every city in `europe_cities_1801.csv`. Wittenberg is in the file under `city_jjk == "WITTENBERG"`.
2. Join the printing and population data from `europe_press_1500.csv` and construct the treatment indicator as in Week 2. Report your row counts.
3. Estimate the first stage. Use the same sample and specification as the Week 2 first stage, with distance from Wittenberg in place of distance from Mainz, so adoption on distance controlling for log city population in 1500. Estimate it three times, keeping only cities within a distance cutoff of Wittenberg each time. Choose the three cutoffs yourself, for example 500, 750 and 1,000 km. Report the coefficient, the standard error, and the sample size for each.
4. **One paragraph.** Does relevance hold? Answer the question the numbers support. A conclusion that the instrument fails is a complete answer if you show why.
5. **One more paragraph.** Suppose relevance had held. Would you believe the exclusion restriction? Name one specific channel, other than printing, by which distance from Wittenberg could affect city growth from 1500 to 1600, the Week 2 outcome.

Audit half. Five errors

The file `code/exercise1_audit.R` arrives with `git pull`. Run it from the repository root, as you run the weekly scripts. It reads like the code a language model writes from a one-sentence description of a task, and five errors of kinds you have met this term were planted in it.

The script is meant to reproduce the Week 2 first stage and reduced form for European cities, distance from Mainz against adoption of a press by 1500 and against growth from 1500 to 1600, and to report the area of each country as a size measure.

It runs from start to finish in a fresh session without errors, and the numbers it prints look plausible.

It contains five errors. Some change the estimates and some change what the script reports. Find them. For each, write down

1. the line, or the lines,
2. what the code actually does, as against what it was meant to do,
3. why nothing complained,
4. the fix,
5. the check that would have caught it before it reached a table.

The fifth point is worth as much as the other four together.

Two notes. Running the script will not tell you what is wrong, because none of the errors raises a message. Read the code. Every error in it is of a kind you have met this term, in a lecture, in a live demo, or in your own code.

The AI use note

Half a page at most. What you asked for, which tools, what came back, what you changed, and what you checked before you signed it. Include anything a model wrote that you did not trust, and say why.

There is no penalty for using AI heavily and no credit for using it lightly. The grade is for the work and for the checking. The note itself carries no marks, and it is required.

Fall break falls between distribution and the due date. The exercise should take about one evening.